Gen AI and Agentic AI in Production

Week 2
Day 1

4 Week Journey to Prod-Grade

Week 4: Agentic AI in Production

- Multi-Agent Architecture
- Capstone: Agents on AWS
- Capstone: Frontend
- Observability Monitoring Security
- Agentic AI Platforms

Week 3: Azure, GCP, Data Engineering, Vectors, MCP

- Cyber AI on Azure with MCP
- Cyber AI on GCP with MCP
- Amazon SageMaker
- Data Ingest S3 Vectors
- Data Pipes with MCP

Week 2: AI Platform Engineering on AWS

- AWS Architecture
- The Digital Twin Mk 2
- Amazon Bedrock
- Terraform
- Github Actions

Week 1: SaaS Generative AI in Production

- First Gen AI Deployment
- Frontend & Backend
- Auth & Subscription
- Healthcare SaaS App
- Deploy to AWS

So much happened!

1. We built a frontend using **NextJS** in **React**, with **Typescript**, with **Pages Router** (client-side), **Tailwind**
2. We built a backend using **FastAPI**
3. We used **Clerk** for user authentication and for subscription plans
4. We packaged into a **Docker container** and tested
5. Then we deployed the container to **AWS App Runner!**

oh and yes again..

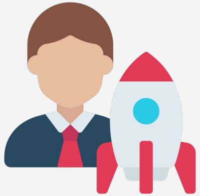
..it **called an LLM and streamed results**... 😂

ASSIGNMENT: Beef up the app! Make it commercially meaningful. Deploy it to Vercel and to AWS to practice both.

Now let's go in, check costs, clean up infrastructure

What skills have you acquired

Entrepreneur



You can deploy an AI web app using Vercel with user authentication and subscription

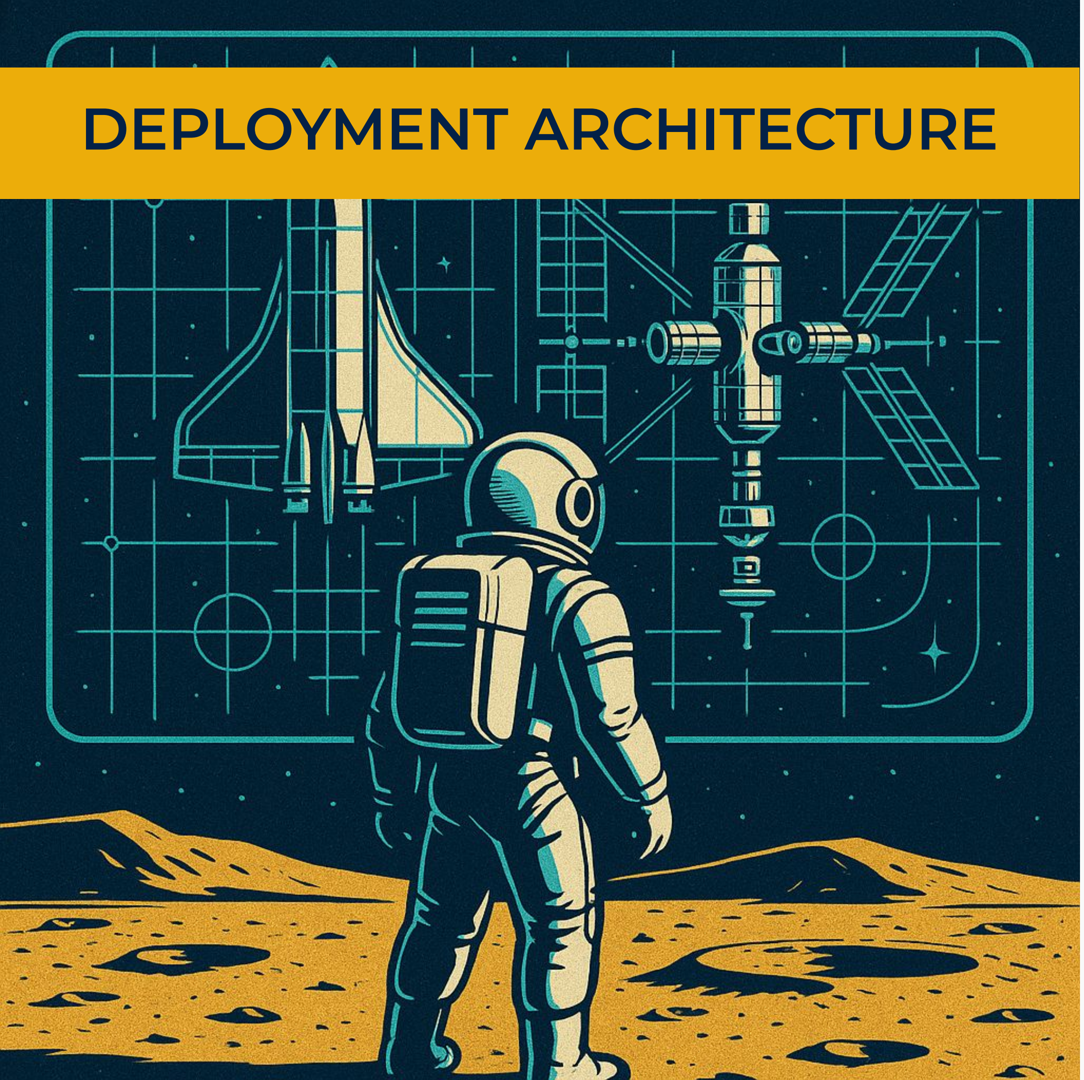
Enterprise

You can set up AWS with IAM then deploy a container app to AWS App Runner via ECR

And I have a gigantic week in store

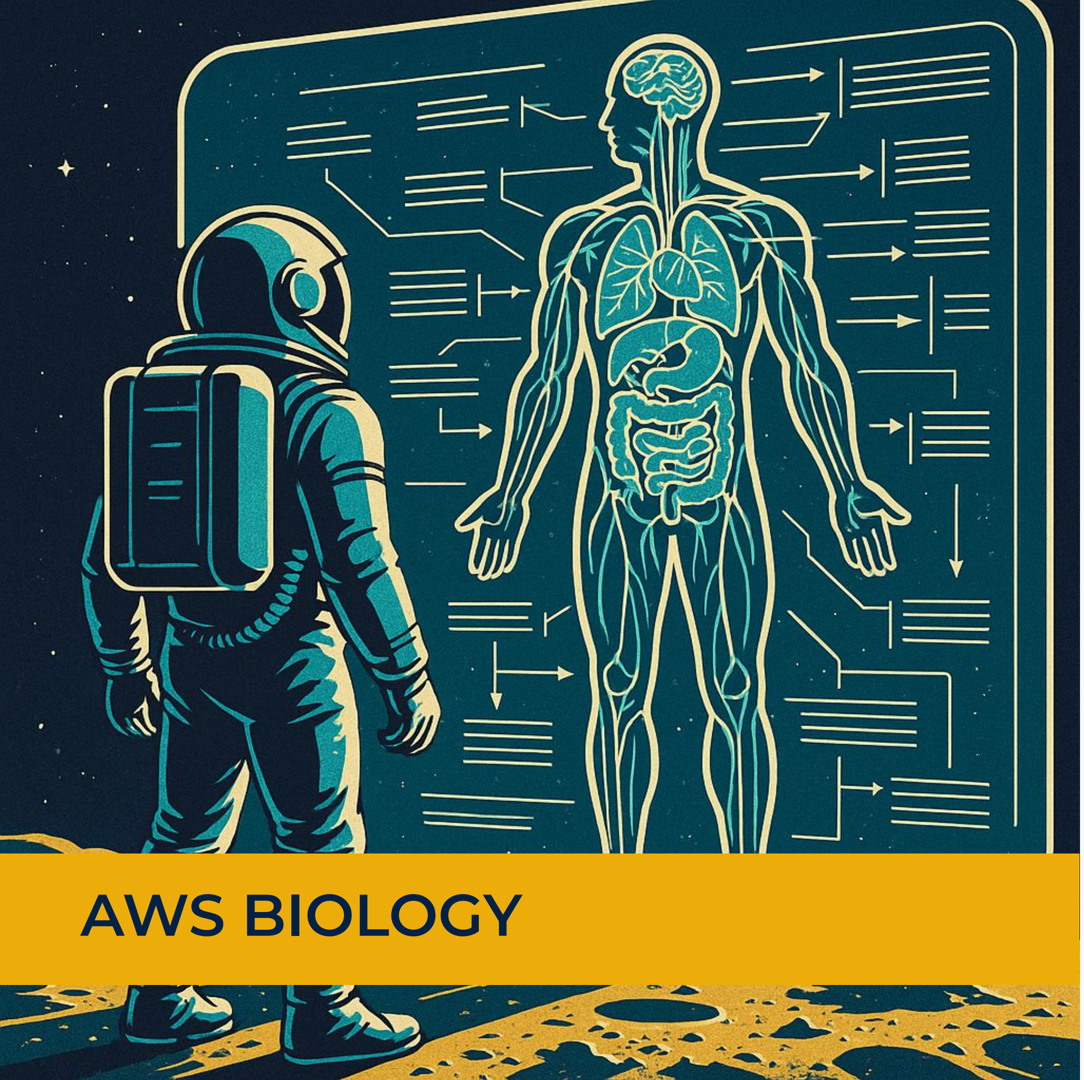
Build and deploy an AI App on AWS
Lambda, Bedrock, S3, API Gateway, CloudFront, Route 53, and more!
Terraform to create infrastructure with code; Github Actions for CICD
For Enterprise: this is core expertise
For Entrepreneur?
I'd argue that this is essential learning for you, too

DEPLOYMENT ARCHITECTURE



Cloud Deployment Archetypes

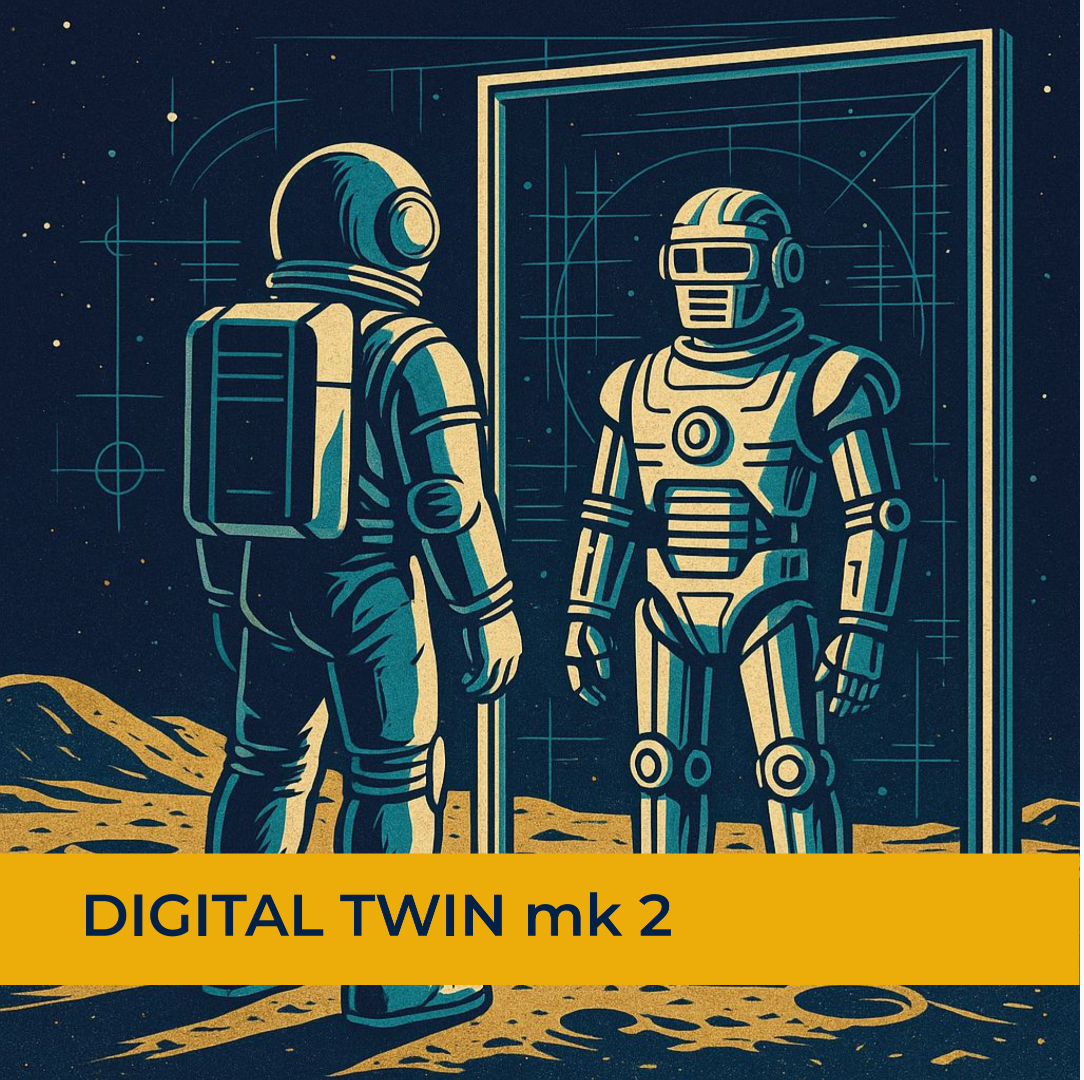
EC2	**Traditional Cloud Servers - IaaS** Where everything started; rent a server and install everything
Vercel Beanstalk	**Platform as a Service - PaaS** Just bring you code; the deployment is handled for you
App Runner	**Container as a Service - CaaS** You provide an app in a container; the service takes care of the rest
ECS EKS	**Container Orchestration** Run your own fleet of containers and manage how they scale
Lambda	**Serverless Functions - FaaS** Upload individual tiny functions and pay per request



AWS BIOLOGY

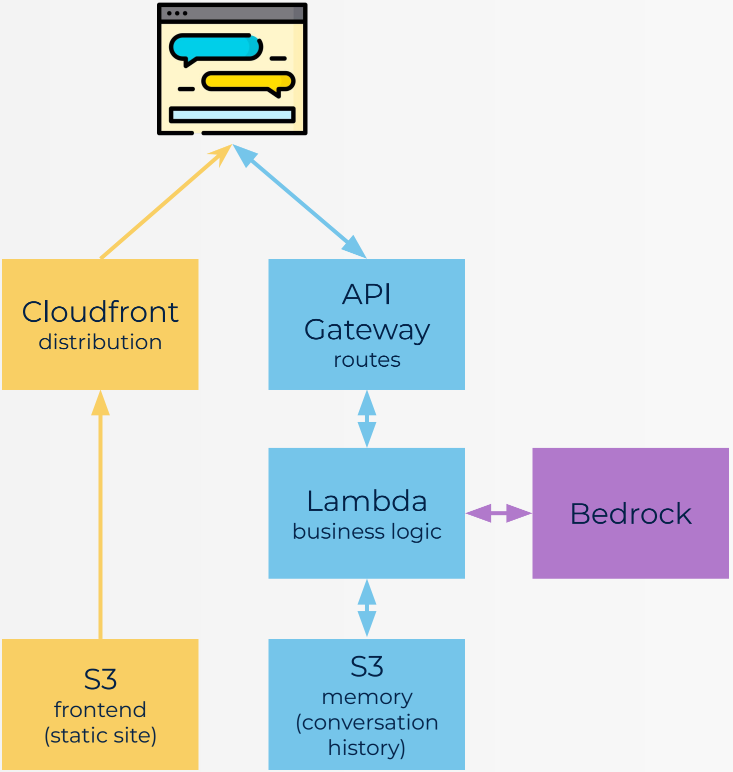
Components this week

S3	**Amazon S3: Simple Storage Service** Like a shared drive in the cloud, organized in “buckets”
Lambda	**AWS Lambda** Individual functions on the cloud; pay for CPU time
CloudFront	**Amazon CloudFront** A content delivery network (CDN) for quickly serving static content
API Gateway	**Amazon API Gateway** Create, manage and scale APIs
Bedrock	**Amazon Bedrock** Quickly build Gen AI applications by connecting to LLMs



DIGITAL TWIN mk 2

Deployment Architecture



AND AGAIN!

Entire week is one intense environment setup
Follow along after the videos
Read the guides, check for updates
Use Cursor Agents / Claude Code:

- Tell them to read the guide
- Show them the errors, get them to check
- Trust nothing, validate everything!

And.. I'm ALWAYS here for questions!!

TO THE LAB!

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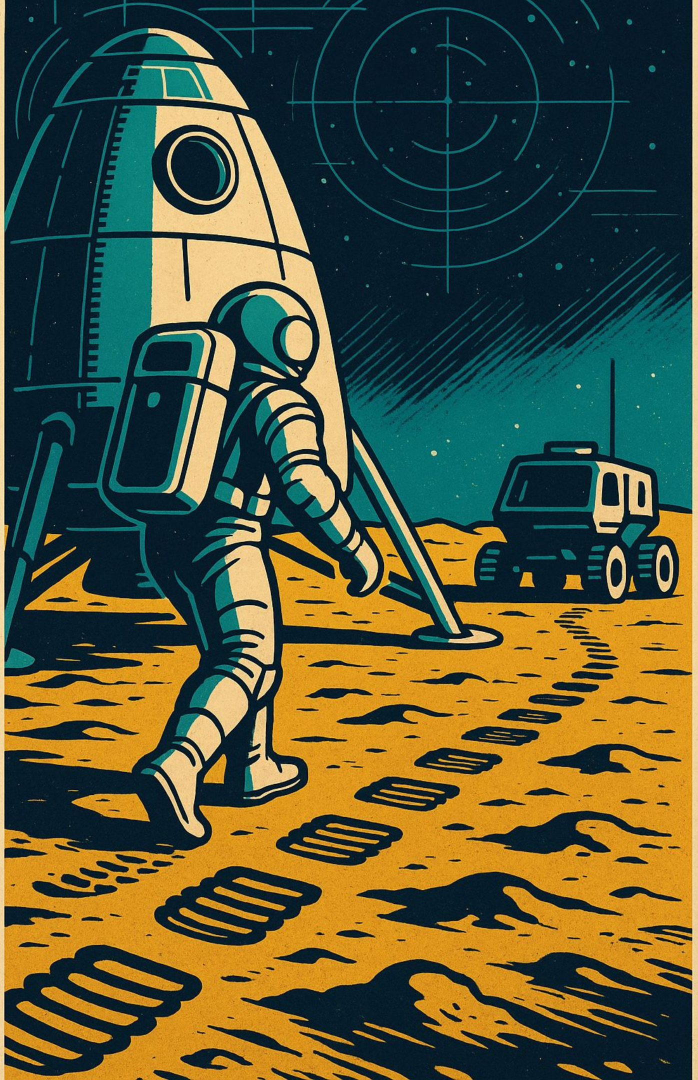

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Complete